

Review of Fundamental Trigonometric Identities

What is an Identity

DEFINITION: Identity

In mathematics, an identity is any equation that holds true for **all** values of the variable, provided those values are from the **domain**.

The Fundamental Trigonometric Identities

We first learned about the fundamental trigonometric identities early on in this workbook. As a refresher, complete the table of identities below from memory.

PROPERTY: Ratio and Reciprocal Identities

Let θ be any value for which the following functions are defined.

Reciprocal Identities	$\sin \theta = \frac{1}{\csc \theta} \qquad \csc \theta = \frac{1}{\sin \theta}$ $\cos \theta = \frac{1}{\sec \theta} \qquad \sec \theta = \frac{1}{\cos \theta}$ $\tan \theta = \frac{1}{\cot \theta} \qquad \cot \theta = \frac{1}{\tan \theta}$
Ratio Identities	$\tan \theta = \frac{\sin \theta}{\cos \theta} \qquad \cot \theta = \frac{\cos \theta}{\sin \theta}$
Pythagorean Identities	$\sin^2 \theta + \cos^2 \theta = 1$ $\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta} \rightarrow 1 + \cot^2 \theta = \csc^2 \theta$ $\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta} \rightarrow \tan^2 \theta + 1 = \sec^2 \theta$

EXAMPLE 1

Let θ be an angle such that $\cos(\theta) > 0$ and $\sin(\theta) = -\frac{2}{3}$. Find $\cos(\theta)$ and $\tan(\theta)$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - (-2/3)^2}$$

$$= \sqrt{1 - 4/9}$$

$$= \sqrt{5/9}$$

$$= \frac{\sqrt{5}}{3}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{-2/3}{\sqrt{5}/3}$$

$$= -\frac{2}{\sqrt{5}}$$

$$= -\frac{2\sqrt{5}}{5}$$

Even and Odd Identities

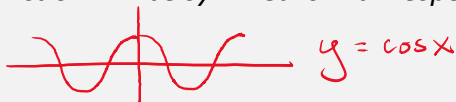
The even and odd identities result from the definition of even and odd functions.

DEFINITION: Even and Odd Functions

The function f is said to be an even function if...

$$\underline{f(-x)} = \underline{f(x)}, \text{ for every } x \text{ in the domain of } f.$$

Note the graph of an even function will be symmetric with respect to the y-axis.



Example: $y = x^2$



$$(-1)^2 = (1)^2 \checkmark$$

The function f is said to be an odd function if...

$$\underline{f(-x)} = \underline{-f(x)}, \text{ for every } x \text{ in the domain of } f.$$

Note the graph of an odd function will be symmetric with respect to the origin.



Example: $y = x^3$



$$(-1)^3 = -(1)^3 \checkmark$$

PROPERTY: Even and Odd Identities

Let θ be any value for which the following functions are defined.

$$\sin(-\theta) = -\sin \theta \quad (\text{odd})$$

$$\csc(-\theta) = -\csc(\theta) \quad (\text{odd})$$

$$\cos(-\theta) = \cos \theta \quad (\text{even})$$

$$\sec(-\theta) = \sec(\theta) \quad (\text{even})$$

$$\tan(-\theta) = -\tan \theta \quad (\text{odd})$$

$$\cot(-\theta) = -\cot(\theta) \quad (\text{odd})$$

EXAMPLE 2

Using the even/odd identities for sine and cosine, along with other fundamental identities, verify the remaining four even/odd identities.

$$\sec(-\theta) = \frac{1}{\cos(-\theta)} = \frac{1}{\cos(\theta)} = \sec(\theta)$$

$$\csc(-\theta) = \frac{1}{\sin(-\theta)} = \frac{1}{-\sin(\theta)} = -\csc \theta$$

$$\tan(-\theta) = \frac{\sin(-\theta)}{\cos(-\theta)} = \frac{-\sin(\theta)}{\cos(\theta)} = -\tan(\theta)$$

$$\cot(-\theta) = \frac{\cos(-\theta)}{\sin(-\theta)} = \frac{\cos(\theta)}{-\sin(\theta)} = -\cot(\theta)$$

EXAMPLE 3

Continuing from Example 1, let θ be an angle such that $\cos(\theta) > 0$ and $\sin(\theta) = -\frac{2}{3}$. Find $\csc(-\theta)$, $\cos(-\theta)$ and $\cot(-\theta)$.

$$\csc(-\theta) = -\csc \theta = \frac{-1}{\sin \theta} = \frac{-1}{-2/3} = \frac{3}{2}$$

$$\cos(-\theta) = \cos(\theta) = \frac{\sqrt{5}}{3} \quad (\text{see example 1})$$

$$\cot(-\theta) = -\cot \theta = \frac{-1}{\tan \theta} = \frac{-1}{-2/\sqrt{5}} = \frac{\sqrt{5}}{2}$$

↑ (see example 1)

Useful Applications of the Fundamental Identities

Trigonometric identities are a powerful tool for solving trigonometric equations. One major hurdle with these equations is dealing with statements involving multiple trigonometric functions. Generally, an equation involving a single trigonometric function is preferred over one with multiple functions.

EXAMPLE 4

Write the trigonometric expression, $-2 \sin^2(x) + \cos(x) + 1$, in terms of a single trigonometric function and factor.

$$\begin{aligned} -2 \sin^2 x + \cos x + 1 &= -2(1 - \cos^2 x) + \cos x + 1 \\ &= -2 + 2\cos^2 x + \cos x + 1 \\ &= 2\cos^2 x + \cos x - 1 \\ \text{let } u &= \cos x &= 2u^2 + u - 1 \\ &= (2u - 1)(u + 1) \\ &= (2\cos x - 1)(\cos x + 1) \end{aligned}$$

EXAMPLE 5

It is known that $\theta \in QI$ and that θ satisfies the equation below. Find θ using the fundamental identities.

$$\begin{aligned} \frac{\sin \theta \cot \theta}{\csc^2 \theta (1 - \cos^2 \theta)} &= \frac{\sqrt{2}}{2} \\ \frac{\cancel{\sin \theta} \frac{\cos \theta}{\cancel{\sin \theta}}}{\frac{1}{\cancel{\sin^2 \theta}} (\cancel{\sin^2 \theta})} &= \frac{\sqrt{2}}{2} \\ \cos \theta &= \frac{\sqrt{2}}{2}, \quad \theta \in QI \\ \theta &= \frac{\pi}{4} \end{aligned}$$

